

Compare cuboid volumes

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Cuboid A is $8 \times 6 \times 5$ cm. Cuboid B is $10 \times 6 \times 4$ cm. Compare their volumes. Copy or complete the first step.

2. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

3. A solid has 5 layers of 24 unit cubes. What is its volume?

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes. Copy or complete the first step.
Answer Both are 240 cm³.
2. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
Answer 240 cubes.
3. A solid has 5 layers of 24 unit cubes. What is its volume?
Answer 120 cubic units.
4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

Compare cuboid volumes

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Cuboid A is $8 \times 6 \times 5$ cm. Cuboid B is $10 \times 6 \times 4$ cm. Compare their volumes.

6. Check this result using an inverse, estimate or known fact: A solid has 5 layers of 24 unit cubes. What is its volume?

2. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

7. Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

3. A solid has 5 layers of 24 unit cubes. What is its volume?

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.

4. Solve this independently and show every stage: Cuboid A is $8 \times 6 \times 5$ cm. Cuboid B is $10 \times 6 \times 4$ cm. Compare their volumes.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

10. Write and solve a similar question to: A solid has 5 layers of 24 unit cubes. What is its volume?

Teacher answers and checking prompts

1. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
Answer Both are 240 cm³.
2. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
Answer 240 cubes.
3. A solid has 5 layers of 24 unit cubes. What is its volume?
Answer 120 cubic units.
4. Solve this independently and show every stage: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
Answer Both are 240 cm³.
5. Use a second method or representation for: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
Answer Expected result: 240 cubes. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A solid has 5 layers of 24 unit cubes. What is its volume?
Answer Expected result: 120 cubic units. A valid check must be shown.
7. Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer 420 cm³. The explanation must show why the method works.
8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A solid has 5 layers of 24 unit cubes. What is its volume?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Compare cuboid volumes

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

6. Create a new example that tests the same idea as: A solid has 5 layers of 24 unit cubes. What is its volume?

2. Prove the answer to this question, rather than only calculating it: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 120 cubic units.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer 420 cm³. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
Answer Expected result: Both are 240 cm³. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
Answer Expected result: 240 cubes. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 120 cubic units.
Answer One valid reconstruction is: A solid has 5 layers of 24 unit cubes. What is its volume? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.
6. Create a new example that tests the same idea as: A solid has 5 layers of 24 unit cubes. What is its volume?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.